QUESTIONS 2

1. Consider the function $f(x) = -x^3 - \cos(x)$.

   a) Use Newton's method with $p_0 = -1$ to find p_2. Could $p_0 = 0$?
   b) Use Secant method with $p_0 = -1$ and $p_1 = 0$ to find p_3.

2. a) Show that for any positive integer k, the sequence defined by $p_n = \frac{1}{n^k}$ converges linearly to $p = 0$.

   b) Show that the sequence defined by $p_n = 10^{-n^k}$ does not converge to $p = 0$ quadratically.